Subarta Ghosh

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PROFESSIONAL SUMMARY

Full Stack Developer skilled in building scalable web applications using Java, Spring Boot, FastAPI, React, REST APIs, and SQL. Experienced in developing backend systems, implementing caching and asynchronous processing, and integrating AI-powered features into modern applications. Strong foundation in Data Structures & Algorithms, OOPs, DBMS, and full-stack development.

EDUCATION

Master of Computer Applications (MCA) – CGPA: 9.27/10 2025–2027
Vellore Institute of Technology, Vellore

B.Sc. (Hons) Computer Science 2022–2025
University of Calcutta, Kolkata

Senior Secondary (CBSE - PCM) 2021–2022

Secondary (ICSE) 2019–2020

TECHNICAL SKILLS

- **Programming Languages:** Java, SQL, Python, JavaScript, C
- **Frontend Technologies:** React.js, HTML5, CSS3, JavaScript
- **Backend & API Development:** Spring Boot 3, REST APIs, Node.js, Express.js, FastAPI
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Tools & Platforms:** Git, Docker, GitHub Actions, Render, Vercel
- **Distributed Systems & Caching:** Apache Kafka, Redis, Caffeine, Event-Driven Architecture
- **Monitoring & Performance:** Prometheus, Grafana, Micrometer, Resilience4j, Bucket4j
- **AI & NLP:** RAG, FAISS, NLP Pipelines, TF-IDF, NMF
- **Core CS Concepts:** Data Structures & Algorithms, OOPs, DBMS, Operating Systems

PROJECTS

Asphalt Prep – Adaptive Learning SaaS Platform Aug 2025 – Jan 2026
Spring Boot 3, PostgreSQL, Redis, Caffeine Live Demo: ai-practice-platform.vercel.app
Aiven Kafka, Resilience4j, Bucket4j, Docker Link: github.com/subh-ghosh/ai-practice-platform

- Developed a scalable full stack learning platform using Spring Boot and React with modular backend architecture.
- Implemented multi-level caching (Redis + Caffeine), reducing mean latency from **196.6ms to 76.4ms** (**61.2% faster**) and P95 latency from **614.9ms to 116.4ms** (**81.1% improvement**) across **22 endpoints**; improved throughput from **14.23 to 15.70 req/s**.
- Applied Bucket4j token-bucket rate limiting; during stress testing blocked **14,687 requests** with a **96.39% block rate**.
- Integrated Resilience4j patterns (circuit breaker, retry, fallback), reducing stress-path latency from **2226ms to 77ms**.
- Introduced asynchronous event processing with Kafka, reducing HTTP failure rate from **1.46% to 0.28%**.
- Integrated **LLM-based features using Grok API** to generate adaptive learning content and dynamic responses, enhancing personalization within the platform.
- Instrumented backend using Actuator, Micrometer, Prometheus, and Grafana for real-time monitoring.
- Deployed backend on Render and frontend on Vercel using Dockerized environments.

Autonomous Fire Detection and Response System Feb 2025 – Jun 2025
Embedded Systems, C++, IoT Link: github.com/subh-ghosh/arduino-based-automated-fire-fighting-robot

- Built an ESP32/Arduino-based fire detection robot using multi-flame sensors and servo-controlled suppression.
- Applied Gaussian-based probabilistic classification replacing static threshold detection.
- Modeled stopping distance using Linear Regression and integrated Blynk IoT alert monitoring.

SignalShift – AI-Powered Customer Feedback Analytics Platform Feb 2026 – Present
Python, scikit-learn, FastAPI, React, NLP, OpenAI API, FAISS Link: github.com/subh-ghosh/signalshift

- Built a scalable backend system using **FastAPI** to process and analyze **191,000+ Google Play Store reviews**, enabling API-driven data pipelines and structured processing workflows.
- Implemented a machine learning pipeline using Logistic Regression and TF-IDF Vectorization (Bigrams), achieving **87.6% accuracy** and a **0.914 F1-score** on test data.
- Integrated **LLM-based summarization and insight extraction** to generate concise summaries and actionable insights from large-scale unstructured text data.
- Designed a **retrieval-augmented (RAG-ready) architecture** using text chunking and embedding-based preprocessing to support semantic search and contextual querying.
- Experimented with **vector similarity search using FAISS** for efficient retrieval of relevant feedback segments.
- Applied **NMF Topic Modeling** to categorize feedback into **12 recurring product issue categories**, enabling prioritization of product improvements.
- Built REST APIs to expose analytics and AI-powered insights to frontend applications.
- Ensured model robustness using **5-fold cross-validation**, achieving a low performance variance of **0.0013** across folds.

ACHIEVEMENTS & CERTIFICATIONS

- Secured Global Rank 840 in **TCS CodeVita Season 13** Certificate Link
- Certified for **SQL (Advanced)** – HackerRank Certificate Link
- Solved 150+ Data Structures & Algorithms problems on coding platforms.